



New Relic for Students

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Getting Started with NRQL and Dashboards

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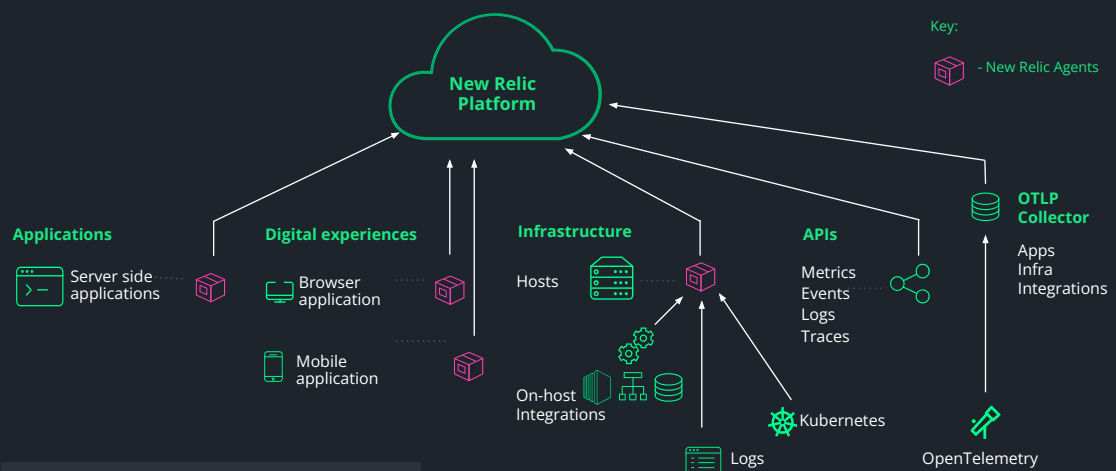
Agenda

- 01 Data types and Data Explorer
- 02 NRQL
- 03 Dashboard fundamentals
- 04 Chart types
- 05 Advanced dashboard techniques

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New Relic Platform



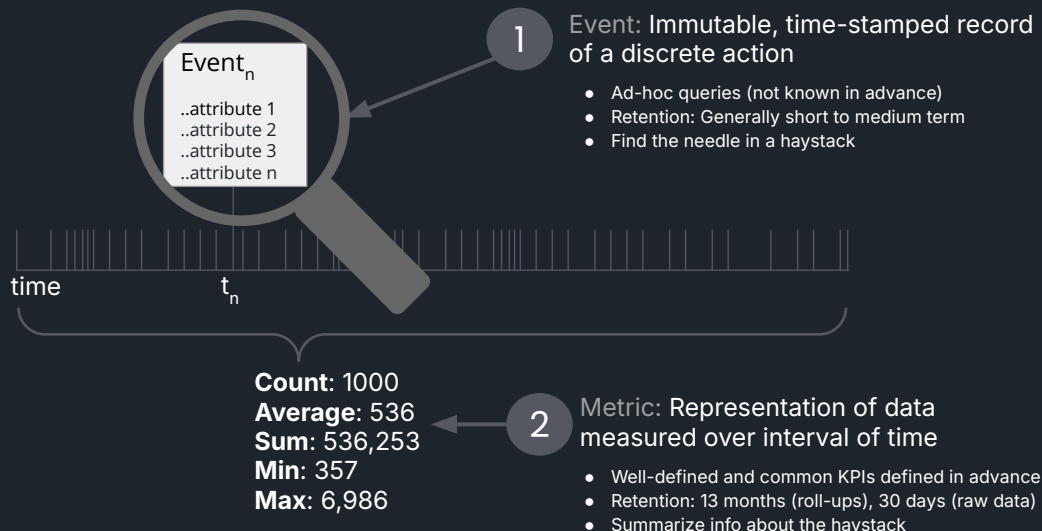
Resources

- [Course: Instrumentation guide](#)

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Metrics vs. Events

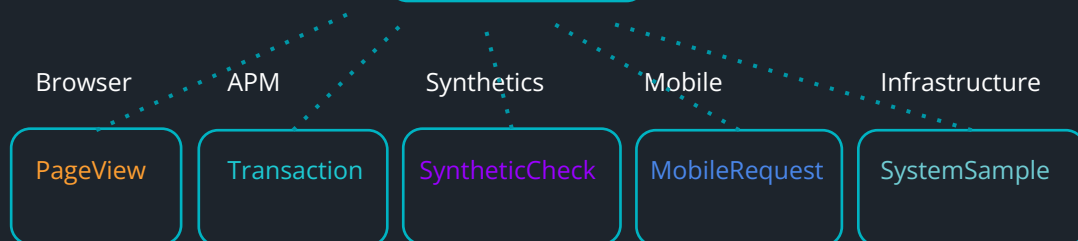


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Common event types

`SELECT function(attribute) FROM EventType WHERE condition FACET group by`



`SELECT count(*) FROM PageView FACET PageUrl SINCE 1 day ago`

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Lab: Data Explorer

- 1 Build these queries by selecting the following options:
 - Event type: **Transaction** > **Count**, **name** > **FACET**
This shows throughput for different Transactions
 - Event type: **Transaction** > **Count**, **appName** > **FACET**
This shows throughput for different Applications
 - Try a different chart type (Bar chart, Pie chart, etc.)
 - Event type: **Transaction**, **duration** > **max**, **name** > **FACET**
This shows slowest Transactions
 - Event type: **PageView**, **duration** > **max**, **pageUrl** > **FACET**
This shows slowest Pages. Try a Bar chart and filter to a pageUrl by clicking one from below
- 2 Try **PageView** > **Raw data** and explore the attributes of the PageView event type

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NRQL syntax

SELECT ... FROM

`SELECT` function(attribute) `FROM` EventType `WHERE` condition `FACET` group by

```
SELECT count(*) FROM Transaction WHERE appName = 'Frontend' FACET name
FROM Transaction SELECT count(*) WHERE appName = 'Frontend' FACET name
```

5 mins

Lab: Explore your data

Try these queries

Select Query Your Data

1

```
SHOW EVENT TYPES
```

Which event types are reporting?

2

```
SELECT keyset() FROM Transaction
```

Explore the attributes. Do you think any of them are custom / business attributes?

3

```
SELECT count(*) FROM Transaction
```

Aggregate the data to retrieve a single value

4

```
SELECT count(*) FROM Transaction FACET appName
```

Select DISTINCT values (GROUP BY) using FACET

Look up default event types and attributes in the [New Relic data dictionary](#)

NOTE: You can't FACET until you aggregate

☐ `SELECT * FROM`
☒ `SELECT count(*)`

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5 mins

Lab: Query Builder

NRQL mode

1

Add a new chart and type one of these queries in the query builder and explore the results
 Choose *Table* view and add one your dashboard

- `SELECT * FROM Transaction`
- `SELECT * FROM PageView`
- `SELECT * FROM SystemSample`

2

Aggregate the data with a count(*) function, and group it using the FACET clause
 Add this chart to your dashboard, for example:

- `SELECT count(*) FROM Transaction FACET appName`

3

Try the same query but try different FACET clauses (for example: name, httpResponseCode, etc)

- `SELECT count(*) FROM Transaction FACET httpResponseCode`

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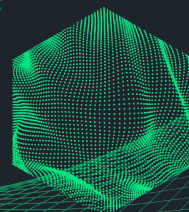
Lab: More advanced NRQL queries

Try some of these queries in the NRU Training account

Use these event types:

- 1 PageView
 - What is the worst (**max**) page load **duration** grouped (**FACET**) by `pageUrl`, `countryCode`, `userAgentName`
- 2 Transaction
 - **Count** `http.statusCodes` WHERE `http.statusCode > '300'` and group the results by `appName` and `http.statusCode`
- 3 Transaction
 - Create a table of average response time (**duration**) labelled AS 'Response time', and average **databaseDuration** AS 'Database time', grouped (**FACET**) by `name` and `request.method`
- 4 PageView
 - Using `count(*)`, create a pie chart showing the breakdown of different browsers (**userAgentName**) and **deviceType**
- 5 SystemSample
 - Show average **cpuPercent** by `hostname`

Monitoring Infrastructure and Backend Services



Agenda

- 01 New Relic APM overview
- 02 Lab: Explore the service summary page
- 03 Lab: Troubleshoot an APM performance issue
- 04 Lab: Using Logs
- 05 Lab: Troubleshoot an Infrastructure issue

Lab: Explore the service summary page

The Service Summary page in APM is a set of curated charts to help you diagnose issues and follow links to delve deeper into related metrics, and troubleshoot the problem. Explore this page for a service and answer the questions below:

1. What is the significance of the four "score cards" at top of the page?
(Hint: Click the question-mark icons above each widget)
2. How can you change the time range displayed on the charts?
3. How can you view the data for individual instances of the application?
4. How can you see what other services or databases it is connected to?
5. How can you view the slowest transactions?
6. Extra credit: What is a transaction trace?

Lab: Troubleshoot an APM performance issue

Find the **Ad Service** in the **Demogorgon Prod** account. Set the time window to the past 60 minutes. Notice that there are several response-time spikes in the past hour

1. Based on the colors on the *Web transactions time* chart, where is most of the time being spent?
2. Which transaction(s) are responsible for the response-time spike?
3. Select the most time-consuming transaction and look at the transaction details. Do you notice anything on that page that may explain why the transaction is taking a long time?
4. Scroll further down the transaction detail page and select a transaction trace with a duration greater than 60,000 ms. Looking at the trace, what do you think caused the transaction to take so long?

Lab: Explore Logs

1. Select the **Demogorgon Prod** account. From New Relic's main menu, select *Logs*
2. How can you see logs from the past 6 hours?
3. Filter the list to show only logs with a level of ERROR. How many logs are there?
4. Add a column to the results table to display the log level and entity.name of each log
5. How can you change your filter query to show only logs that contain a value for entity.name?
6. How could you save this view to quickly recall it later?
7. Click on a log entry and see its attributes. How can you easily share the details of a log entry with your team? What if they don't have access to your New Relic account?

Lab: Troubleshoot an Infrastructure issue

Scenario: The New Relic Pet Clinic application experienced a sudden increase in response time. Use Connected Infrastructure and APM to try to identify the problem.

Select the *New Relic Pet Clinic* application in the **NRU Training** account. Set the time picker to 24 hours, and locate a response-time spike (there should be one each day between 06:10 and 06:50 UTC).



Lab: Troubleshoot an Infrastructure issue

Use Connected Infrastructure and APM to answer the following questions:

- Were there any application deployments around the time of the incident?
- Was there an increase in application throughput that might explain the increased response time?
- Did response time increase across all hosts (containers), or only certain ones?
- Select an affected container and view the data for the associated host (*Hint: Related entities > See full map*) Do you notice anything unusual about that host's data (*Hint: Check host processes*)